The Blaise Pascal Quantum Challenge
Call for Ideas

Deliverable template

Quantum-Enhanced Optimisation for Prosumer Energy Grids

User guide for this template to be filled and joined to your participation.

- *Please respect the three parts to be completed*
- *Please note that the items suggested inside each part are indications to help you understand what we expect, they are not mandatory*
- *Please insert directly inside the end of your deliverable any relevant document supporting your ideas*

PART 1: Sustainability Impact Assessment

Objective of this section: Demonstrate the measurable and scalable benefits of your project in addressing sustainability challenges.

What to include:

Impact on the challenge selected:

- Describe how your solution aligns with one of the Five Challenges and contributes to the corresponding UN SDGs.
- Anticipate briefly the possibilities for scaling your solution beyond its initial application.

Quantifiable Metrics:

- Specify measurable outcomes, such as energy savings, carbon footprint reduction, improvements in resource efficiency or societal benefits.
- Propose metrics for evaluating success, such as reduction in energy consumption, time savings enabled by quantum AI, social or economic impact on target communities.

(Attach supporting evidence, such as existing studies, data, or examples, if available.)

Impact on the Challenge Selected

Our solution addresses the challenge of integrating renewable energy sources into modern power grids by leveraging Quantum AI to optimise energy distribution. This directly supports Sustainable Development Goal 7 (SDG 7)—ensuring access to affordable, reliable, and sustainable energy.

Environmental Impact

Traditional power grids struggle to efficiently manage decentralised renewable energy sources particularly renewable prosumers (energy systems that both produce as well as consumer energy such as home solar panel systems), leading to energy waste and inefficiencies. By integrating Quantum AI, surplus energy from prosumer households can be rapidly allocated to areas with high demand, significantly reducing reliance on fossil fuels and cutting carbon emissions. This accelerates the transition to clean energy and improves grid resilience against fluctuations in renewable energy generation. Research supports that quantum optimisation can enhance energy grid performance (Blenninger et al., 2024; Mastroianni et al., 2022)

Social Impact

Energy inequality is a growing issue, with fluctuating costs disproportionately affecting lower-income households. By optimising energy allocation, our solution enhances grid stability and ensures fair distribution of renewable power. This reduces energy costs for consumers, contributing to affordable

electricity and alleviating the cost-of-living crisis. Furthermore, by encouraging decentralised renewable energy adoption like home based solar panel systems, it empowers communities to become self-sufficient and less dependent on centralised power providers. For example there are case studies of residential energy management systems optimising renewable usage for community benefit (Kelepouris et al., 2024).

Economic Impact

Quantum AI improves energy forecasting and real-time optimisation, minimising inefficiencies and reducing market volatility. More accurate forecasting leads to stable electricity prices, this in turn creates greater business confidence and investment in renewable energy infrastructure. Additionally as mentioned before widespread adoption of prosumer energy grids promotes energy independence, allowing households and businesses to generate, store, and sell excess electricity. This creates new economic opportunities, lowers long-term electricity costs, and supports sustainable economic growth. Studies have shown the feasibility of prosumer-based models in creating economic and operational efficiencies (Bektaş and Bhuiyan, 2024).

Scalability and Future Applications

Our solution is highly scalable and adaptable to various grid systems, from localised community microgrids to large-scale national power networks. As the number of prosumer participants grows, Quantum AI ensures that energy distribution remains efficient and resilient, making it a future-proof approach for sustainable energy management. The use of Advanced Metering Infrastructure (AMI), already outlined in research in conjunction with quantum computing, further enhances the system's ability to adapt to different energy market structures and evolving grid demands (Boddu et al., 2023).

Quantifiable Metrics

To evaluate the success and impact of our solution, we propose the following key performance indicators (KPIs):

- **Energy Efficiency Gains**: Percentage reduction in energy waste due to improved load balancing (Blenninger et al., 2024).
- **Carbon Footprint Reduction**: Measurable decrease in CO_2 emissions from reduced reliance on fossil fuel-generated electricity (Mastroianni, Scarcello and Settino, 2022)
- **Cost Savings**: Reduction in energy prices due to improved energy forecasting and optimised grid management (Bektaş and Bhuiyan, 2024).

- **Renewable Energy Utilisation**: Increase in the proportion of energy supplied by renewable prosumer sources within the grid (Kelepouris et al., 2024).
- **Time Savings**: Reduction in energy redistribution time compared to classical optimisation methods (Boddu et al., 2023).
- **Grid Stability**: Measurable improvements in voltage stability, storage management, and reliability (Kelepouris et al., 2024).

By integrating Quantum AI with existing grid technologies, this solution enhances sustainability, reduces costs, and increases energy security, making it a possible pathway to a clean, equitable, and efficient energy future.

Bektaş, E. and Bhuiyan, M.R.A., 2024. Optimization based energy management approach for prosumers located in small-scale microgrids. In: 3rd International Karatekin Science and Technology Conference, Çankırı, Türkiye, November 2024. pp. 114-119.

Blenninger, J., Bucher, D., Cortiana, G., Ghosh, K., Mohseni, N., Nüßlein, J., O' Meara, C., Porawski, D. and Wimmer, B., 2024. Q-GRID: Quantum Optimization for the Future Energy Grid. *KI-Künstliche Intelligenz*, pp.1-11.

Boddu, B., Krishna, M.H., Kumar, N., Singh, N. and Lafta, A.M., 2023, December. Quantum Computing Approaches to Optimal Power Flow in Complex Grid Networks. In *2023 International Conference on Power Energy, Environment & Intelligent Control (PEEIC)* (pp. 752-756). IEEE.

Kelepouris, N.S., Kelepouris, D., Bouhouras, A.S. and Christoforidis, G.C., 2024. Prosumer energy management under controllable power transactions with the grid. *IEEE Transactions on Industry Applications*.

Mastroianni, C., Scarcello, L. and Settino, J., 2022, September. Quantum computing approach for energy optimization in a prosumer community. In *2022 IEEE Intl Conf on Dependable, Autonomic and Secure Computing, Intl Conf on Pervasive Intelligence and Computing, Intl Conf on Cloud and Big Data Computing, Intl Conf on Cyber Science and Technology Congress (DASC/PiCom/CBDCom/CyberSciTech)* (pp. 1-7). IEEE.

PART 2: Technical Feasibility of the Solution

Objective of this section: Provide a detailed explanation of how your project leverages quantum computing and its feasibility.

What to include:

General Approach and Strategy Proposed:

- Explain the overall strategy for solving the identified problem.
- Detail how the problem will be solved using AI and quantum computing.
- (Optional) Give a draft comparison between the sustainability benefits (eg. savings in carbon emissions and energy consumption) vs. the footprint of your quantum AI stack (eg. Carbon emissions and energy consumption of the Pasqal quantum as well as the non-quantum parts of the AI system involved), you could use this methodology as a possible guide.

Quantum Computing Integration:

- Explain how quantum computing supports your solution.
- Highlight compatibility with Pasqal's neutral atom technology or other quantum architectures.

Mathematical Problem Formulation:

- Clearly define the problem in mathematical terms, including:
- Key variables, parameters, and equations/models.
- Assumptions or simplifications made.
- Computational complexity and how quantum computing addresses it.

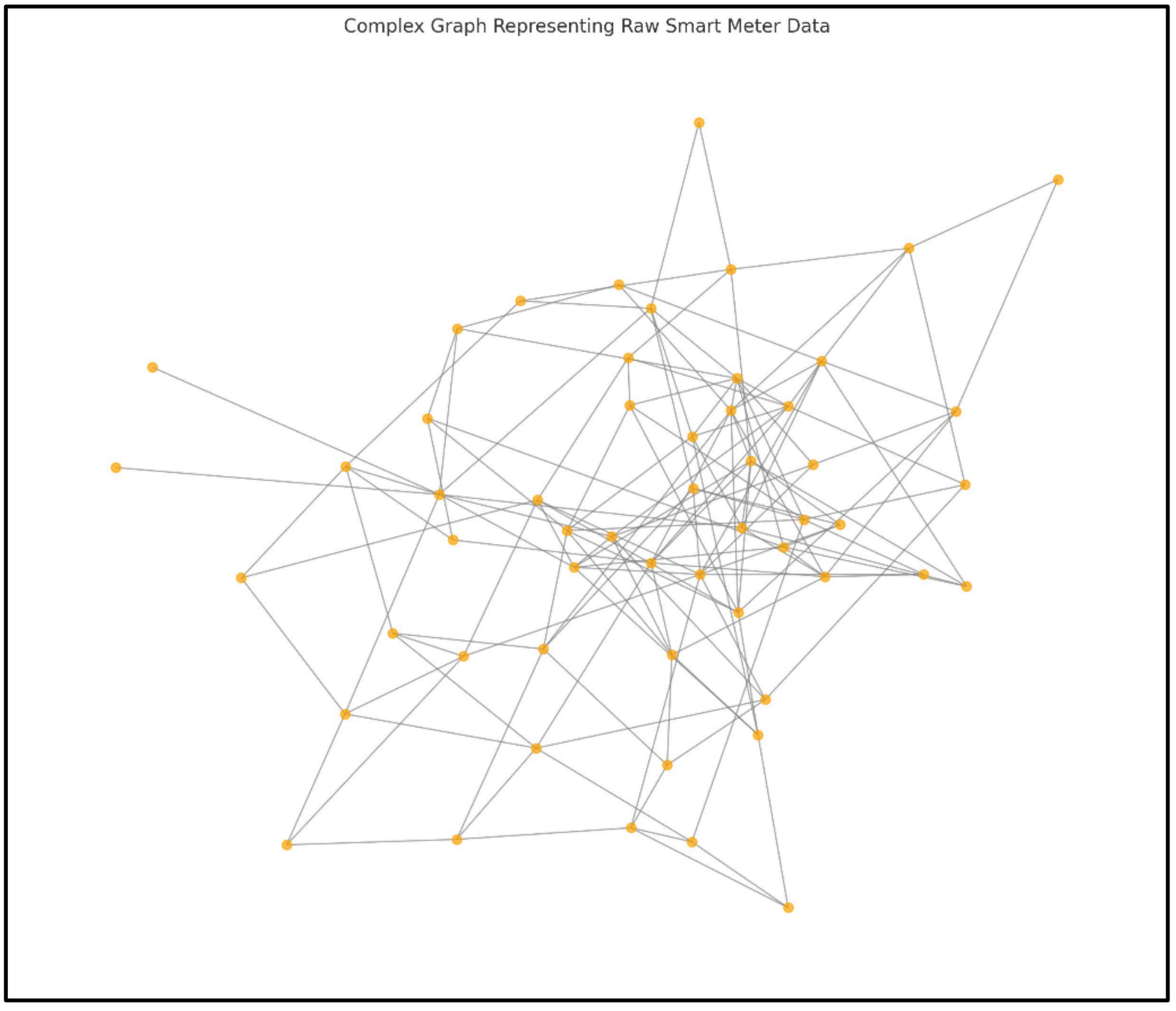
(Attach technical schematics, models, or preliminary simulations, if relevant.)

General Approach and Strategy Proposed

Our solution aims to optimise energy distribution in prosumer grids by leveraging Quantum AI to enhance the efficiency and effectiveness of energy management. A primary challenge lies in handling the vast amounts of real-time data generated by Advanced Metering Infrastructure (AMI) systems and managing decentralised renewable energy sources. Traditional computational methods struggle to process complex, interconnected graphs derived from these datasets, making it difficult to optimise energy flow across the grid.

To address this, we propose using Pasqal's quantum computing technology to perform graph optimisation and energy flow tasks that are computationally intensive for classical systems. Moreover, we integrate several time series forecasting models (such as Long Short-Term Memory (LSTM)

networks, ARIMA, and Autoencoders) to provide key input forecasts for the quantum-enhanced optimisation, making the process more accurate and adaptable.



Visualisation of energy data generated by smart meters, This graph illustrates the energy flow in a traditional grid with prosumer data, highlighting the complexity that traditional methods face when processing this data.

The forecasting models will play a critical role by generating predictions about energy consumption, production patterns from renewable sources, and grid demand. These predictions will then be processed through the quantum computing framework, allowing us to optimise energy flow and distribution in real-time.

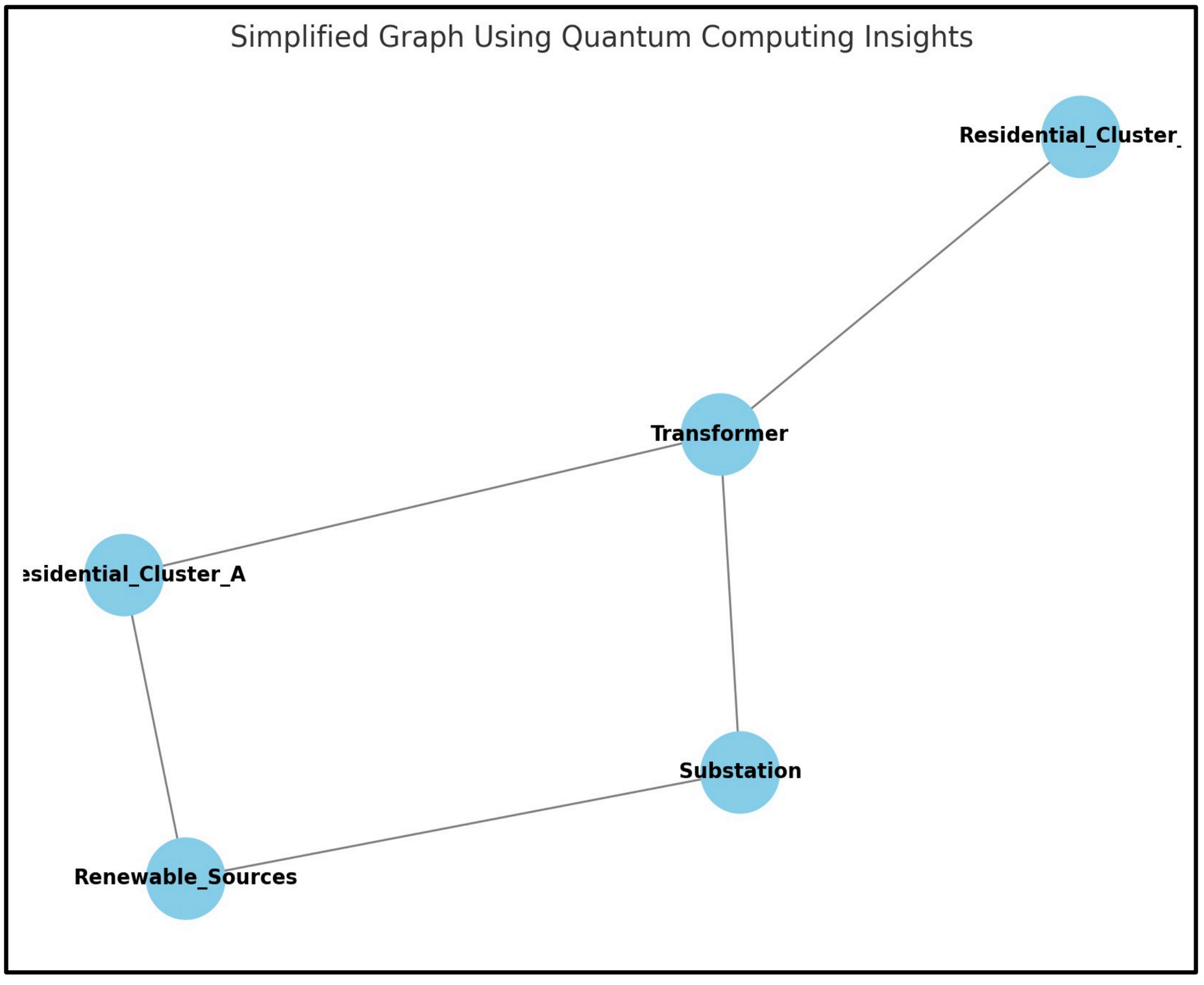
For long-term predictions (such as seasonal energy demand), LSTM networks will be used, while ARIMA models will forecast short-term trends (e.g., daily or weekly demand). Additionally, Autoencoders will help detect anomalies in real-time consumption patterns, enabling us to identify faults or unusual events in the grid that might impact the overall system.

Quantum Computing Integration

Quantum computing supports our solution by enabling efficient processing of large datasets that model energy distribution, consumption patterns, and grid behaviour. Traditional computational methods face challenges in managing the exponential complexity involved in processing these large, interconnected data structures. By using Pasqal's neutral atom quantum processors, our solution can map energy flow problems onto quantum states, performing parallel operations that speed up the problem-solving process (Bodduet al., 2023).

The time series models (LSTM, ARIMA, Autoencoders) will be integrated as input to the quantum system, enhancing its accuracy. These AI-driven models will predict future energy demand, supply from renewable sources, and detect anomalies in the grid. The quantum computing component will then use

these predictions to optimise energy flow, reducing transmission losses, balancing supply and demand, and improving overall grid stability. Once the quantum processor handles the complex calculations, AI can use the insights generated to fine-tune decisions related to resource allocation, grid upgrades, and predictive maintenance (Boddu et al., 2023).



Quantum-processed visualisation of the same energy data, illustrating how quantum processors optimise energy flow and identify patterns that are more actionable than traditional methods. This

figure shows how Pasqal's quantum processor optimises the energy flow graph, with reduced energy losses and better load balancing.

Technology Used: Quantum Analog Computing

Pasqal's quantum analog computing approach allows the direct simulation of physical models that mirror the networked behaviour of energy grids. Quantum processors' ability to explore and optimise energy flow within a high-dimensional Hilbert space makes them particularly effective for managing the large datasets generated by smart meters in AMI systems (Kelepouris et al., 2024).

The LSTM models will predict long-term consumption trends (e.g., weekly or seasonal energy demands), and ARIMA will forecast short-term demand fluctuations (e.g., daily, hourly). These predictions will serve as inputs into the quantum optimisation process, allowing us to compute optimal energy flow configurations in real time.

Autoencoders will help in detecting anomalies or faults in the energy grid by analysing historical data and identifying any unusual patterns in energy consumption. Once quantum processors handle the initial graph optimisation, AI can utilise the processed data for further tasks:

- Training predictive models to identify long-term trends in energy consumption.
- Enhancing decision-making for grid maintenance and upgrades.
- Creating user-friendly interfaces and visualisations to present actionable insights for utilities.

This collaboration between quantum computing and AI ensures that energy flows are optimised, and decision-making becomes faster and more accurate, leading to enhanced grid efficiency.

Mathematical Problem Formulation

To mathematically formulate the problem, we define the energy flow as a function of various variables, such as energy consumption, grid connectivity, and production from prosumer sources. The optimisation problem can be represented as a graph where nodes correspond to energy sources (e.g., prosumer homes, substations) and edges represent the energy flow between nodes. The objective is to minimise energy losses and reduce transmission inefficiencies while balancing supply and demand.

That is minimise:

$$\sum_{(i,j)\in E} c_{ij} x_{ij}$$

Where:

- c_ij represents the cost or energy loss associated with the energy transfer between nodes i and j.
- x_ij is the energy flow between nodes i and j.
- E is the set of edges in the grid graph.

Assumptions include linear power flow equations and a simplified grid structure for computational feasibility. However, these can be extended to more complex models as needed. Quantum computing accelerates the search for the optimal solution by enabling parallel computation over vast solution spaces, thus reducing the computational complexity significantly compared to classical methods.

Computational Complexity and Quantum Solutions

Classical optimization of energy flow in large-scale grids involves solving NP-hard problems, where the time to find a solution grows exponentially with the size of the grid. Quantum computing, particularly using Pasqal's neutral atom processors, reduces this complexity by exploiting quantum parallelism, allowing us to solve these problems much faster (Blenninger et al., 2024). The use of quantum computing also addresses the challenge of scaling these solutions across larger grids and more complex network structures, ensuring that our system remains efficient as the prosumer grid expands.

The integration of LSTM, ARIMA, and Autoencoder models ensures that the data fed into the quantum system is already optimised for long-term predictions, short-term forecasting, and anomaly detection, allowing quantum computing to focus solely on solving the complex optimisation tasks.

Time Series Forecasting Inputs

Optimising energy grid systems requires integrating multiple approaches to handle both short-term fluctuations and long-term trends while ensuring grid stability and performance. Several recent studies have explored these challenges by combining short-term forecasting, long-term forecasting, and anomaly detection methods.

For short-term predictions, classical time series methods such as ARIMA have been widely used to forecast immediate energy consumption and generation. However, to better manage grid stability

during unpredictable energy fluctuations, LSTM (Long Short-Term Memory) networks are being employed for long-term forecasting, capturing more complex temporal dependencies and long-term trends in energy supply and demand (Sharma et al., 2023). This combination ensures that both immediate and future energy needs are considered for efficient grid management.

Anomaly detection plays a crucial role in maintaining the health and reliability of the grid. Autoencoders and similar unsupervised learning techniques are often used to identify irregular patterns in the data, such as energy theft, cyberattacks, or other operational issues (Zhang et al., 2021). The integration of such methods allows for the quick identification and mitigation of anomalies that could disrupt energy distribution.

Recent advancements have demonstrated that these techniques can be effectively combined to optimise energy grids. For example, a somewhat recent review emphasizes how machine learning, including autoencoders, aids in both predictive modelling and anomaly detection to ensure optimal grid performance while reducing inefficiencies (Singh & Gupta, 2023).

Furthermore, studies highlight the potential for advanced machine learning techniques to improve grid efficiency, using a mix of forecasting and anomaly detection to create a more resilient energy system (Kumar et al., 2023).

These combined approaches demonstrate the vital role of AI and machine learning techniques in ensuring energy grid systems are both efficient and resilient. The integration of LSTM for long-term forecasting, ARIMA for short-term predictions, and autoencoders for anomaly detection forms a robust framework for optimising energy distribution, especially when used as inputs in a quantum computing optimisation algorithm which theoretically is more efficient than the classical computing equivalent.

QAOA Integration

The Quantum Approximate Optimization Algorithm (QAOA) plays a pivotal role in this solution by addressing the combinatorial optimisation tasks involved in energy flow and distribution. In the case of energy grids, QAOA optimises the allocation of energy from decentralised prosumer sources to high-demand areas by efficiently navigating the vast solution space. By leveraging quantum parallelism and quantum gates, QAOA can rapidly evaluate different configurations of energy flows, helping us achieve optimal distribution faster than classical methods (Farhi et al., 2014).

The application of QAOA in optimising the energy grid’ s performance goes beyond simply reducing

energy losses—it also accelerates fault detection, load balancing, and predictive maintenance. By solving these optimisation tasks more efficiently, we can achieve greater reliability and stability in the grid (Farhi et al., 2014).

Blenninger, J., Bucher, D., Cortiana, G., Ghosh, K., Mohseni, N., Nüßlein, J., O'Meara, C., Porawski, D. & Wimmer, B., 2024. Q-GRID: Quantum Optimization for the Future Energy Grid. *KI-Künstliche Intelligenz*, pp. 1-11.

Boddu, B., Krishna, M.H., Kumar, N., Singh, N. & Lafta, A.M., 2023. Quantum Computing Approaches to Optimal Power Flow in Complex Grid Networks. In *2023 International Conference on Power Energy, Environment & Intelligent Control (PEEIC)*, IEEE, pp. 752-756.

Farhi, E., Goldstone, J., Gutmann, S., et al., 2014. A Quantum Approximate Optimization Algorithm. *arXiv:1411.4028*.

Kelepouris, N.S., Kelepouris, D., Bouhouras, A.S. & Christoforidis, G.C., 2024. Prosumer Energy Management Under Controllable Power Transactions with the Grid. *IEEE Transactions on Industry Applications*.

Kumar, R., Singh, P. & Verma, A., 2023. Energy grid optimization using deep machine learning. *Energy and AI*, 13, p. 100258. Available at: https://doi.org/10.1016/j.egyai.2023.100258.

Mastroianni, C., Scarcello, L. & Settino, J., 2022. Quantum Computing Approach for Energy Optimization in a Prosumer Community. In *2022 IEEE Intl Conf on Dependable, Autonomic and Secure Computing*, pp. 1-7.

Sharma, S., Raj, A. & Kumar, M., 2023. Time series forecasting of energy demand using LSTM networks. *International Journal of Energy Research*, 47(8), pp. 3432–3445. Available at: https://doi.org/10.1002/er.8322.

Singh, R. & Gupta, P., 2023. AI-empowered methods for smart energy consumption. *Renewable and Sustainable Energy Reviews*, 155, p. 111885. Available at: https://doi.org/10.1016/j.rser.2022.111885.

Zhang, L., Chen, Z. & Li, Y., 2021. Time series anomaly detection for smart grids: A survey. *IEEE Transactions on Smart Grid*, 12(4), pp. 2715–2726. Available at:

https://doi.org/10.1109/TSG.2020.3032305.

PART 3: Innovation and Creativity

Objective of this section: Highlight the originality and disruptive potential of your quantum AI solution.

What to include:
State of the Art:

- Position the solution within existing research and technologies.
- Justify the novelty of the approach, describe whether your project involves:
    - A new quantum AI algorithm.
    - An innovative application of an existing algorithm.
    - An adaptation of known methods to a novel use case.

(Attach scientific papers and any documentation available to support this subsection)

Disruption and Uniqueness:

- Explain how your solution offers an original contribution to the field.
- Highlight specific areas where it disrupts or advances beyond current practices.

Alignment with Quantum AI Trends:

- Position your project within the broader quantum AI landscape.
- Discuss its potential to contribute to advancements in both sustainability and quantum AI.

State of the art

The current landscape of energy grid optimisation primarily relies on Advanced Metering Infrastructure (AMI) systems. These systems employ smart meters to collect data on energy consumption, transmitting this information to utility companies via communication networks. While AMI systems have proven effective in providing real-time insights and enabling basic optimisations, they face challenges when it comes to handling vast amounts of data, managing interconnected elements, and ensuring scalability as grids grow larger and more complex particularly as energy grids try and convert to renewable energy sources and thus increase in their number of prosumers. To address these challenges we propose our solution of using quantum computing to optimise the energy grid.

At the core of our solution lies the integration of quantum computing with AI-based forecasting methods, a novel approach within the specific context of prosumer energy grids. While quantum computing and AI time series forecasting and analysis are separate fields, combining these technologies for grid optimisation represents an innovative application. Quantum algorithms, such as Quantum Approximate Optimisation Algorithms (QAOA), have been used in other domains to optimise

combinatorial problems. However, their direct application to AMI systems is a pioneering development. Our approach repurposes existing quantum optimisation methods to address a critical challenge in the energy sector: optimising energy distribution and management across dynamic, large-scale grids in real time.

Our system uses quantum computing to perform highly complex optimisation tasks more efficiently than classical computing systems, especially as grid sizes scale. Classical methods typically rely on polynomial or exponential time complexity to solve optimisation problems (e.g., $O(n^3)$ or $O(2^n)$), while quantum algorithms like QAOA exhibit logarithmic scaling, enabling much faster solutions for larger problems. This scalability advantage of quantum computing is crucial when applied to energy grids that continuously grow in size and complexity.

Additionally, we enhance our quantum AI system with time series forecasting. The inclusion of time series analysis ensures the grid is dynamically optimised, with real-time predictions, as well as future projections, guiding energy distribution and fault detection.

Disruption and Uniqueness

Our solution is disruptive because it introduces quantum computing as a scalable alternative to classical optimisation methods in the energy sector. As grids become larger, classical optimisation approaches struggle to handle the computational burden associated with the large-scale, interconnected data that AMI systems generate. This is particularly true for dynamic grids in prosumer energy systems, where energy flows are complex and constantly evolving. Classical algorithms typically become inefficient or infeasible as the grid expands, especially when faced with millions of variables such as prosumers, consumers, energy storage systems, and renewable energy sources.

Quantum computing, on the other hand, offers a significant advantage. Quantum Approximate Optimisation Algorithms (QAOA), when applied to combinatorial optimisation problems, offer a logarithmic scaling of time complexity, meaning the number of operations required grows much slower than in classical methods. For example, for a grid with 10,000 variables, classical computing might require billions of operations, whereas quantum computing might require just 20 operations (refer to our analysis in the Appendix where we compare the scale up of classical optimisation algorithms with QAOA). This dramatic reduction in computational complexity becomes even more pronounced as the grid size increases, enabling real-time optimisation of energy systems even in highly dynamic environments.

What sets our approach apart is not just the quantum advantage, but also how we combine quantum computing with AI-driven forecasting. By leveraging time series analysis, we can forecast energy demand in both the short term and long term, ensuring the grid remains adaptive and optimally managed over time. Furthermore, the integration of quantum algorithms with AI for anomaly detection means the system can proactively identify issues in the grid before they escalate into larger problems, providing a level of predictive power that classical systems cannot offer.

The novelty lies in applying quantum optimisation directly to the highly dynamic, real-time energy data generated by AMI systems. Unlike traditional systems that might struggle with scalability or efficiency, our solution offers an entirely new approach to grid optimisation, capable of handling increasing data complexity and ensuring that energy flows are optimally managed, even as systems expand.

Alignment with Quantum AI Trends

This project places itself at the forefront of the quantum AI revolution, contributing to the rapidly evolving field of quantum machine learning and quantum optimisation. While quantum computing has gained significant attention in research for its potential applications in finance, logistics, and materials science, its practical application in energy systems—specifically within prosumer grids—remains largely unexplored. Our work pushes the boundaries of quantum AI by applying these technologies in the context of real-time energy management.

This solution is aligned with ongoing trends in both sustainability and quantum AI development. By optimising energy grids, particularly those incorporating renewable energy sources and prosumer models, our solution has the potential to improve the efficiency of energy systems, reduce waste, and enhance grid reliability. This is crucial for supporting global sustainability goals, where optimising energy use and reducing carbon emissions are key priorities. At the same time, the integration of quantum computing and AI signals a significant leap forward in the commercialisation of quantum technologies, opening new pathways for quantum AI applications in industries that deal with complex optimisation challenges.

By introducing time series forecasting within this framework, we also align our work with the growing demand for predictive analytics in the energy sector, offering an innovative approach to both short-term and long-term planning. This makes the system not only a tool for optimisation but also a strategic asset in managing energy supply and demand across future grid systems.

Pasqal

APPENDIXES: Please feel free to add any relevant document here

APPENDIX A: Comparing Classical and Quantum Scaling:

As the grid size increases, it's essential to compare how classical algorithms and quantum algorithms scale in terms of computational time. Below is an expanded analysis incorporating classical graph optimisation algorithms as well as time complexity comparisons with quantum approaches.

Classical optimisation methods often rely on polynomial or exponential time complexity to solve combinatorial problems. For large-scale grids, the time complexity increases significantly as the number of variables grows

Some Common Classical Graph Optimisation Algorithms

Refer to Cormen et al. (2009) for the time complexities of the different algorithms.

Dijkstra's Algorithm:

Time Complexity:

- **Basic:** $O(V^2)$ with an adjacency matrix.
- **Optimised:** $O((V+E)logV)$ with a priority queue.
- **Scaling for large grids:**
  - As the number of nodes (V) in the grid increases (which corresponds to the number of prosumers, consumers, or nodes in a smart grid), the time complexity increases quadratically for the basic implementation or logarithmically for the optimised version with priority queues.
- **Example:** For a grid with 10,000 nodes and 50,000 edges:
  - Basic Implementation: $O(10{,}000^2) = 100$ million operations.
  - Optimised Implementation: $O((10{,}000 + 50{,}000)\log 10{,}000) = O(60{,}000 * 4)$ $\approx 240{,}000$ operations.

Prim's Algorithm:

- **Time Complexity:** $O(ElogV)$

- **Scaling for large grids:**
  - In grids with a high number of edges (E), the time complexity is dependent on the edge-to-vertex ratio. As the number of edges increases, the operations grow linearly with the number of edges, but logarithmically with the number of vertices.
- **Example:** For a grid with 10,000 vertices and 100,000 edges:
  - O(100,000 log 10,000) = 400,000 operations.

Kruskal's Algorithm:

- **Time Complexity:** $O(ElogV)$
- **Scaling for large grids:**
  - Similar to Prim's algorithm, Kruskal's scales with the number of edges (E) and vertices (V). The sorting step dominates the complexity.
- **Example:** For a grid with 10,000 vertices and 100,000 edges:
  - O(100,000 log 10,000) = 400,000 operations.

Quantum Algorithms (e.g., Quantum Approximate Optimisation Algorithm, QAOA)

Quantum algorithms, such as QAOA, offer a significant advantage in terms of scaling. These quantum methods can solve combinatorial optimisation problems much faster, with logarithmic or polylogarithmic time complexity (Farhi et al., 2014).

They have a Time Complexity of $O(polylog(n))$ or $O(polylog(n)p)$, where n is the number of variables and p is the number of layers in the quantum circuit.

For example:

- For a grid with 10,000 variables, assuming quantum algorithms scale poly-logarithmically with n, the time complexity might be:

$$Tquantum(10,000) = O(polylog(10,000)p)$$

Since polylog(10,000) ≈ log(10,000) = 4, and assuming 5 iterations (layers), the computational time would be approximately:

$$Tquantum(10,000) \approx 4 \times 5 = 20$$ operations.

- For a grid with 1 million variables:

$$Tquantum(1,000,000) = O(polylog(1,000,000)p)$$

Since polylog(1,000,000) ≈ log(1,000,000) = 6, and assuming 5 iterations (layers), the computational time would be approximately:

$$Tquantum(1,000,000) \approx 6 \times 5 = 30$$ operations

Conclusion: Comparative Scalability

In summary, classical graph optimisation algorithms such as Dijkstra's and Kruskal's algorithms struggle as the grid grows larger due to the inherent limitations in scaling. For a grid with 1 million variables, classical approaches like Linear Programming or Integer Programming could require trillions of operations, making them inefficient for real-time optimisation. Graph-based algorithms such as Dijkstra’ s and Prim's can reduce operations, but their complexity still becomes prohibitive for very large networks.

In contrast, quantum optimisation algorithms, such as QAOA, show promise in solving combinatorial optimisation problems exponentially faster. With time complexities scaling logarithmically, quantum algorithms provide a scalable solution that can efficiently handle real-time energy grid optimisation, even as the number of variables and grid complexity increases.

By integrating quantum optimisation with AI-driven forecasting, our solution can handle the increasing scale of complexities of modern energy grids, making it not only faster but also far more efficient than classical counterparts.

Cormen, T.H., Leiserson, C.E., Rivest, R.L. & Stein, C., 2009. *Introduction to Algorithms* (3rd ed.). Cambridge, MA: MIT Press.

Farhi, E., Goldstone, J., Gutmann, S. et al., 2014. A Quantum Approximate Optimisation Algorithm. *arXiv:1411.4028*. Available at: https://arxiv.org/abs/1411.4028 [Accessed 30 January 2025].